YC1121S

High Performance Low Power BLE 5.4 SoC

Preliminary Datasheet

General Descriptions

The YC1121S is a high performance, low power System-on-Chip (SoC) integrating a Bluetooth® 5.4 compliant 2.4-GHz transceiver, 24 MHz proprietary 32 bit MCU with a RAM of 8 KB and a One-Time Programmable (OTP) memory of 8KB .

The YC1121S supports Bluetooth Low Energy and Bluetooth 5.4 features including high-throughput 2 Mbps, Long Range and the Direction Finding. It can be paired through HCI interface with a more powerful MCU for applications requiring advanced wireless connectivity.

The fully-featured multiprotocol radio, +8 dBm output power, -96.5 dBm sensitivity and extended temperature range of -40 to 125°C makes it suitable for lighting applications.

The YC1121S features built-in USB, proprietary 32-bit MCU clocked at 24 MHz, integrated capless LDOs supporting 1.6-5.5V supply range, making it a perfect microcontroller for cost-sensitive applications.

Key Features

- MCU subsystems
  - 24 MHz 32-bit proprietary MCU for system control and PHY/link layer management
  - AES128 HW encryption
  - Serial wire debug
- Memories
  - 8 KB OTP with internal charge pump
  - 8 KB data RAM (4 KB with retention)
  - 8 KB Patch RAM with retention
- Radio transceiver
  - Bluetooth 5.4/Long Range
  - +8 dBm TX power
  - -96.5 dBm RX sensitivity @ BLE 1 Mbps
  - -93 dBm RX sensitivity @ BLE 2 Mbps
  - -98.5 dBm RX sensitivity S2
  - -103 dBm RX sensitivity S8
  - Integrated balun with single-ended output and direct connection to antenna
  - 5.4 mA RX system current @ BLE 1 Mbps -97 dBm sensitivity (ideal DC-DC converter: 3V→1.5V)
  - 9.5 mA TX system current @ 0 dbm (ideal DC-DC converter: 3V→1.5V)
- Power management
  - Always-On (AON) supply: 1.6~ 5.5V
  - Main supply: 1.5 ~ 5.5V supporting external DCDC through a dedicated wakeup pin
  - Integrated LDOs requiring no external decoupling capacitors
  - 1.2 µA in sleep mode (wake on RTC, no RAM retention)
  - 5µA in sleep mode (wake on RTC,12 KB RAM retention)
- Clock generation
  - Dedicated PLL to support 16M/24M/32M crystals
  - Crystal trimming
  - 24MHz RC oscillator for fast wakeup
  - Low jitter low power 32 KHz RC oscillator
- 9-channel 12-bit ADC
- Digital peripherals
  - Up to 24 GPIOs w/ functions fully multiplexed
  - 8 x PWMs up to 24 Mbps
  - Two-wire master (I^2C compatible) up to 600 kbps
  - 1 x UART(RTS/CTS) with HCI-H5 protocol up to 3.25 Mbps
  - 1 x SPI Master up to 24 Mbps
  - 1-axis Quadrature Decoder
  - Tscan
  - IR
  - 12 Mbps Full Speed USB 2.0
- Audio Codec
  - Mono differential PWM DAC
  - Mono audio ADC with PGA
  - Microphone bias generator
  - ADPCM support
- Temperature range: -40°C to +125°C
- HBM 2KV, CDM 500V

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Applications

- Mouse devices
- Toys
- Lightning applications
- Disposables
- Commercial and industrial applications requiring advanced connectivity

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Revision History

Version	Date	Owner	Note
1.0	25/10/2023	MS.M	Initial version
1.1	22/2/2024	MS.M	Correct incorrect data in Table 3-2

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1 Block Diagram

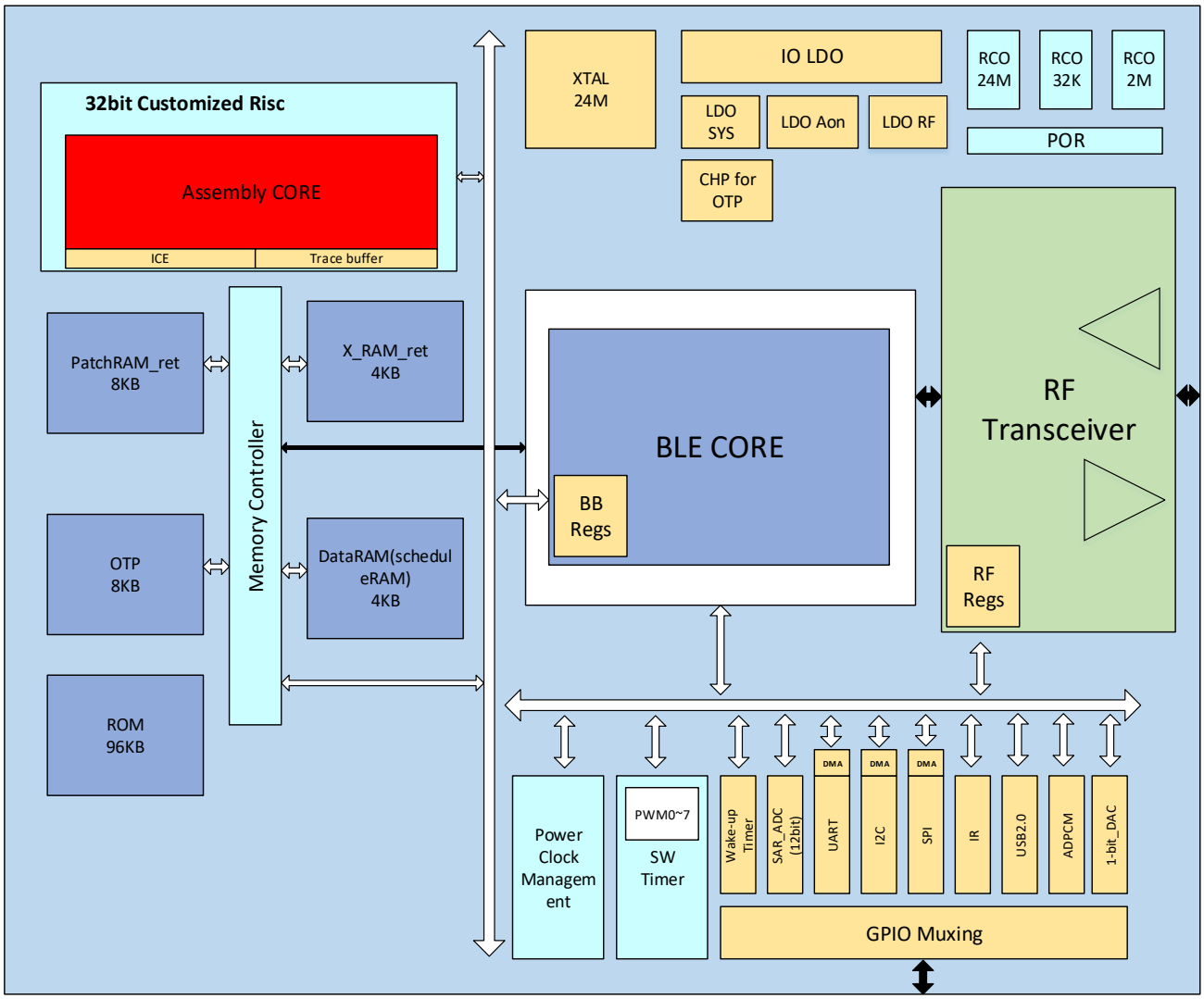

Figure 1-1 Block diagram

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2 Pinout Information

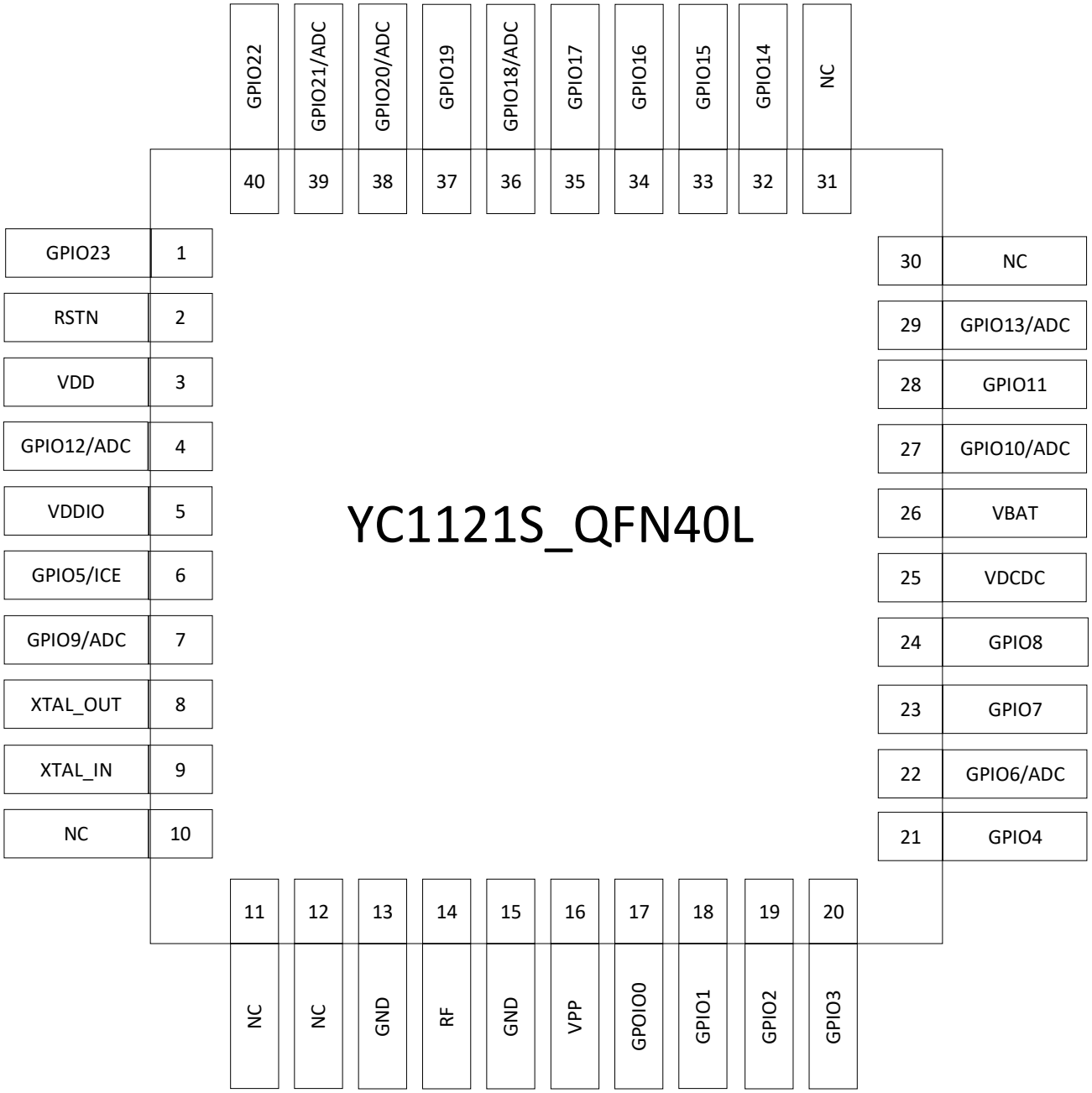

Figure 2-1 Pinout top view (QFN40 package)

Abbreviations:

PWR: Power pin

AIO: Analog IO pin

DIO: Digital IO pin

RF: RF IO pin

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Table 2-1 Pinout Information

Pin Number QFN40	Type	Name	Description
1	DIO	GPIO23	Refer to Table 2-2
2	DIO	RSTN	Global reset, active low. OR gated with internal POR. NC if not needed.
3		VDD	
4	DIO	GPIO12/ADC	Refer to Table 2-2
5		VDDIO	
6	DIO	GPIO5/ICE	Refer to Table 2-2
7	DIO	GPIO9/ADC	Refer to Table 2-2
8	AIO	XTAL_OUT	XTAL port
9	AIO	XTAL_IN	XTAL port, or external reference clock input
10		NC	
11		NC	
12		NC	
13	GND	GND	
14	RF	RF	ANT port
15	GND	GND	
16		VPP	
17	DIO	GPIO0	Refer to Table 2-2
18	DIO	GPIO1	
19	DIO	GPIO2	
20	DIO	GPIO3	
21	DIO	GPIO4	
22	DIO	GPIO6/ADC	
23	DIO	GPIO7	
24	DIO	GPIO8	
25	PWR	VDCDC	Main power input, 1.5~5.5V
26	PWR	VBAT	Always-on power input, 1.6~5.5V
27	DIO	GPIO10/ADC	Refer to Table 2-2
28	DIO	GPIO11	
29	DIO	GPIO13/ADC	
30		NC	
31		NC	
32	DIO	GPIO14	Refer to Table 2-2
33	DIO	GPIO15	Refer to Table 2-2
34	DIO	GPIO16/USBDP	USB port
35	DIO	GPIO17/USBDN	USB port
36	DIO	GPIO18/ADC	Refer to Table 2-2
37	DIO	GPIO19	
38	DIO	GPIO20/ADC	
39	DIO	GPIO21/ADC	
40	DIO	GPIO22	

Table 2-2 GPIO Multiplexing

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Pin Name	boot function	function-analog
GPIO[0]		
GPIO[1]		
GPIO[2]		
GPIO[3]		
GPIO[4]		
GPIO[5]	ICE	
GPIO[6]		saradc [4]
GPIO[7]		
GPIO[8]		
GPIO[9]		saradc [2]
GPIO[10]		saradc [1]
GPIO[11]		
GPIO[12]		saradc [3]
GPIO[13]		saradc [8]
GPIO[14]		
GPIO[15]		
GPIO[16]		
GPIO[17]		
GPIO[18]		saradc [0]
GPIO[19]		
GPIO[20]		saradc [5]
GPIO[21]		saradc [6]
GPIO[22]		saradc [7]
GPIO[23]	EXEN	

3 Specifications

3.1 Recommended Operating Conditions

Table 3-1 Recommended Operation Condition

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Supply voltage for pin VBAT	VBAT		1.6		5.5	V
Supply voltage for pin VDCDC	VDCDC		1.5		5.5	V
Supply voltage for pin VIO	VIO			3.3	3.6	V
Ambient temperature	TA		-40		125	°C

NOTE: the high-voltage(above 4.2V) mode needs to be configured.

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3.2 Power Consumption

Table 3-2 Power Consumption Characteristics

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit	Comment
Sleep							
Current through pin VBAT	IVDCDC_SLEEP	VBAT= 3.0V		1.2		µA	
	IVDCDC_SLEEP_FULL_RET			6		µA	(1)
Current through pin VDCDC	IVDCDC_SLEEP	VDCDC = 1.5V		10		nA	
RX mode 1 Mbps BLE @ -96.5 dBm sensitivity							
Current through pin VBAT	IVBAT_RX	VBAT = 3.0V		2.8		mA	
Current through pin VDCDC	IVDCDC_RX	VDCDC = 1.5V, VDCDC = 3V		5.2		mA	
TX mode P_{out} 0dBm							
Current through pin VBAT	IVBAT_TX	VBAT= 3.0V		2.5		mA	
Current through pin VDCDC	IVDCDC_TX	VDCDC = 1.5V, VDCDC = 3V		14.0		mA	

(1) FULL_RET(Full retention) is 12 KB RAM retention.

3.3 Radio

All parameters are referred to chip port and measured on the condition of VDCDC = 3.3V if not stated otherwise.

Table 3-3 Transmitter Specification

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Frequency range	f_{TX}		2300		2600	MHz
Output power	P_{out}		-20.0		8	dBm
Power control step	P_{step}	For part-to-part power calibrations		3		dB
Spurious emissions (@ 0dBm)	P_{spur}	30 MHz to 1000 MHz		-43.7		dBm
		1 GHz to 12.75 GHz		-31.0		dBm
		47 MHz to 74 MHz		-75		dBm
		87.5 MHz to 108 MHz		-75		dBm
		174 MHz to 230 MHz		-75		dBm
		470 MHz to 862 MHz		-44.0		dBm

Table 3-4 Receiver Specification

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Frequency range	f_{RX}		2300		2600	MHz
Out-of-band blocking	OOB	30 MHz – 2000 MHz	-30			dBm

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		2003 – 2399 MHz	-35			dBm
		2484 – 2997 MHz	-35			dBm
		3000 MHz – 12.75 GHz	-30			dBm
1 Mbps BLE						
RX sensitivity	P_{SENS_1M}	30.8% PER		-96.5		dBm
C/I co-channel	C/I_{CO_1M}	30.8% PER	21	10		dBc
C/I 1 MHz adjacent channel	C/I_{1_1M}	30.8% PER	15	-4		dBc
C/I 2 MHz adjacent channel	C/I_{2_1M}	30.8% PER	-17	-35		dBc
C/I ⩾3 MHz adjacent channel	C/I_{3_1M}	30.8% PER	-27	-37		dBc
C/I image channel	C/I_{im_1M}	30.8% PER	-9	-30		dBc
C/I image channel + 1MHz	C/I_{im+1_1M}	30.8% PER	-15	-37		dBc
Maximum input signal level	$P_{IN_MAX_1M}$	30.8% PER	-10	0.0		dBm
2 Mbps BLE						
RX sensitivity	P_{SENS_1M}	30.8% PER		-93		dBm
C/I co-channel	C/I_{CO_2M}	30.8% PER	21	10		dBc
C/I 2 MHz adjacent channel	C/I_{2_2M}	30.8% PER	15	-4		dBc
C/I 4 MHz adjacent channel	C/I_{4_2M}	30.8% PER	-17	-35		dBc
C/I ⩾6 MHz adjacent channel	C/I_{6_2M}	30.8% PER	-27	-40		dBc
C/I image channel	C/I_{im_2M}	30.8% PER	-9	-30		dBc
C/I image channel + 2MHz	C/I_{im+2_2M}	30.8% PER	-15	-40		dBc
Maximum input signal level	$P_{IN_MAX_2M}$	30.8% PER	-10	0.0		dBm

3.4 24 MHz Crystal Oscillator

Table 3-5 24 MHz Crystal Oscillator Characteristic

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Crystal frequency	f_{XTAL}			24		MHz
Crystal frequency tolerance	Δf_{XTAL}		-20		20	ppm
Load capacitance	$C_{L, INN}$	Programmable via registers		9		pF

Note: the temperature and pressure limitations of BOM(bill of material), including but not limited to the need for high temperature resistant crystals when used at 125 ℃.

3.5 LDO Characteristics

Table 3-6 IO LDO Specification

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Input voltage range	VBAT		1.6	3.3	5.5	V
Output voltage	VOUT_ACTIVE	I_{LOAD}=20 mA, when input voltage below 3.3V, output equals input		3.3		V
	VOUT_SLEEP	I_{LOAD}=100 μA, when input voltage below 3.3V, output equals input		3.3		V
Maximum load current	ILOAD	Active mode@VBAT=5V			150	mA

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3.6 Reset Characteristics

Reset voltage is monitored on pin VBAT_HIGH.

Table 3-7 Reset Characteristics

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Reset voltage threshold	VPOR	rising edge		1.6		V
	VPDR	falling edge		1.44		V
POR stretch time	TPOR			12		mS
PDR stretch time	TPDR			500		µS

3.7 Audio Characteristics

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
DAC PWM Driver			0.5		Watts
PGA gain		-2		36	dB
ADC Signal to Noise Ratio	1kHz 0.5Vrms		72		dB
ADC Full Scale	Vbat=3.3V		1		Vrms

Auxiliary SARADC characteristic (1121sb)

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
Resolution			12		Bit
input range		0		VREF	V
Input resistance	With input buffer	1M			Mohms
	With 1/2 divider		0.45		
Sampling rate				1.5	MHz
ENOB	VREF=3.3V, GPIO path		10		Bit
	VREF=3.3, internal battery path		8		Bit

VREF can be selected from 3.0V/3.3V/1.8V ldo output.

4 Application Schematic

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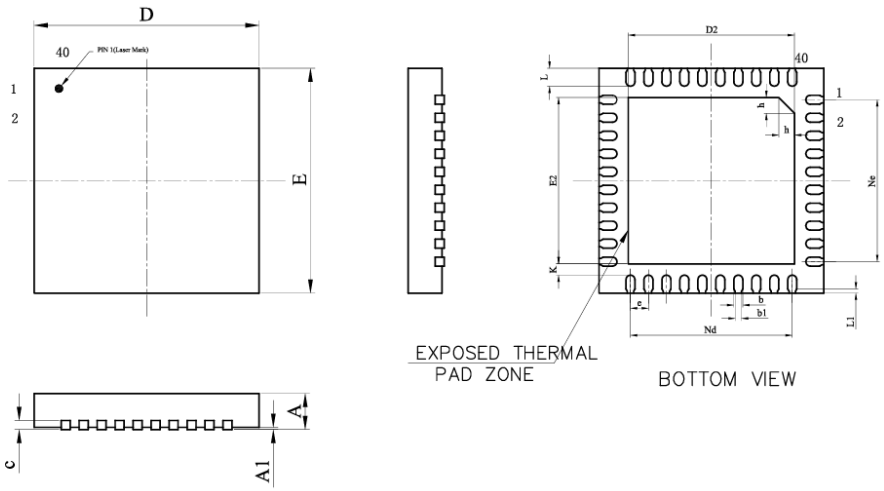
5 Package Information



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.70	0.75	0.80
A1	—	0.02	0.05
b	0.15	0.20	0.25
b1	0.14REF		
c	0.18	0.20	0.25
D	4.90	5.00	5.10
D2	3.60	3.70	3.80
e	0.40BSC		
Nd	3.60BSC		
E	4.90	5.00	5.10
E2	3.60	3.70	3.80
Ne	3.60BSC		
L	0.35	0.40	0.45
L1	0.10REF		
K	0.20	--	--
h	0.30	0.35	0.40
L/F载体尺寸 (mil)	154*154		

Figure 5-1 QFN40 package dimensions

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